Accelerated Math II/III Name_____

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HW #1 - Polynomial Operations & Classification

Date_____ Period____

Write the polynomial in standard form. Name each polynomial by degree and number of terms.

1) $-8x^4 + 5x^3 - 6x^5 - 2x^2 - 1$

2) $2 - 8p^2$

3) $-7 + 10x^4 + 10x$

4) -9

5) $4 + 9n$

6) $2n^3 - 6 - 2n$

Find each product. Write your answer in standard form.

7) $(x + 3)(4x - 1)(3x + 7)$

8) $(x + 1)(x - 1)(x + 2)$

9) $(x + y)(x - y)(x + y)(x - y)$

Perform the indicated operation.

10) $h(n) = n + 5$
$g(n) = n^3 + 2n$
Find $(h + g)(n)$

11) $f(a) = a^2 - 5a$
$g(a) = 3a - 5$
Find $(f - g)(a)$

12) $g(x) = 4x + 1$
$h(x) = x^3 - 3x^2$
Find $(g \cdot h)(x)$

13) $g(n) = n^2 - 1$
$h(n) = -4n + 2$
Find $g(n) \div h(n)$

14) $g(x) = x^2 - 5$
$f(x) = x + 3$
Find $g(10) + f(10)$

15) $g(t) = 3t - 4$
$f(t) = 4t - 5$
Find $g(-2) - f(-2)$

16) $g(x) = x^2 + 5x$
$h(x) = 2x$
Find $g(-7) \cdot h(-7)$

17) $g(t) = -2t^2 - 6t$
$h(t) = 3t + 1$
Find $\left(\frac{g}{h}\right)(6)$

18) $h(x) = -2x - 4$
$g(x) = x^2 - 1$
Find $(h + g)(x^2)$

19) $g(t) = t^2 - t$
$h(t) = 3t$
Find $g(2 - t) + h(2 - t)$

20) $g(x) = x - 3$
$h(x) = x^2 - x$
Find $(g - h)(z - 1)$

21) $f(n) = 2n - 4$
$g(n) = 4n$
Find $f(-3n) - g(-3n)$

22) $h(n) = 3n - 1$
$g(n) = -2n^2 - n$
Find $(h \cdot g)(-3n)$

23) $g(t) = t - 5$
$h(t) = 4t + 5$
Find $g(t^2) \cdot h(t^2)$

24) $g(a) = a^2 - 2a$
$f(a) = -4a + 5$
Find $g(-3x) \div f(-3x)$

25) $h(n) = 4n - 1$
$g(n) = 3n - 4$
Find $h\left(\frac{n}{4}\right) \div g\left(\frac{n}{4}\right)$

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Answers to HW #1 - Polynomial Operations & Classification

1) quintic polynomial with five terms
2) quadratic binomial
3) quartic trinomial
4) constant monomial
5) linear binomial
6) cubic trinomial
7) $12x^3 + 61x^2 + 68x - 21$
8) $x^3 + 2x^2 - x - 1$
9) $x^4 - 2x^2y^2 + y^4$
10) $n^3 + 3n + 5$
11) $a^2 - 8a + 5$
12) $4x^4 - 11x^3 - 3x^2$
13) $\frac{n^2 - 1}{-4n + 2}$
14) 108
15) 3
16) -196
17) $-\frac{108}{19}$
18) $x^4 - 2x^2 - 5$
19) $t^2 - 6t + 8$
20) $-z^2 + 4z - 6$
21) $6n - 4$
22) $162n^3 - 9n^2 - 3n$
23) $4t^4 - 15t^2 - 25$
24) $\frac{9x^2 + 6x}{12x + 5}$
25) $\frac{4n - 4}{-16 + 3n}$

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